

Q1.

 Campus Hiring - Technical 2025 - New Construct Test Timer: 00:40:00

4. Computer Science  < 1 2 3 4 5 6 7 8 9 10 > 

Question 9
Assume a mobile application needs to retrieve a list of all available restaurants from a food delivery service's API. The application should only read the data and not make any changes to the server's resources. Which common HTTP method is most appropriate for this type of data retrieval? Select the correct option from the given choices.


Attempted: 0/14
 Revisit Later

Select an option
☐ POST
☐ PUT
☐ DELETE
☐ GET

Correct answer : GET

Explanation : The GET method is used to retrieve data from a server without modifying the server's resources. Since the application only needs to retrieve a list of available restaurants, GET is the most appropriate HTTP method.

Q2.

 Campus Hiring - Technical 2025 - New Construct ⌚ Test Timer: 00:40:00

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Question 5 Attempted: 0/14
 Revisit Later

Which concept is being used in the given code? Select the correct option from the given choices.

```
1  class A{  
2      int x;  
3      void sum(){  
4          x++;  
5      }  
6      void sum(int a){  
7          x = x + a;  
8      }  
9  }  
10 //code
```



Select an option

☐ Abstraction

☐ Encapsulation

☐ Inheritance

☐ Polymorphism

Correct answer : Polymorphism

Explanation : Defining multiple methods with the same name (sum) but different parameter lists within the same class is method overloading, which is a form of compile-time polymorphism.

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Finish Test

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Computer Science

Attempted: 0/14

Revisit Later

Question 1

What will be the output of the given code when the user clicks a button?

```
1 <body>
2   <button onclick="myFunction(this);">Test onclick attribute!</button>
3   <script type="text/JavaScript">
4     function myFunction(button) {
5       document.write(button.getAttribute("onclick", 1));
6     }
7   </script>
8 </body>
```

Select an option

Clear Response

☐ True

☐ False

☐ myFunction (this);

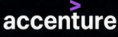
☐ None of the given options

→

Correct answer : myFunction(this);

Explanation : button.getAttribute("onclick") returns the exact literal string value assigned to the HTML attribute, which is "myFunction(this);".

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4. Computer Science

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Clear Response

Question 1

Attempted: 0/14

Revisit Later

As a computer architect, you are designing the control unit of a CPU. You need to identify the data it should fetch the accumulator to the memory. Which of the following best matches this operation?

←

Select an option

☐ The process of transferring data from the accumulator to memory is called a register transfer.

☐ The process of transferring data from the accumulator to memory is called a memory read operation.

☐ The process of transferring data from the accumulator to memory is called a memory write operation.

☐ The process of transferring data from the accumulator to memory is called an instruction fetch.

☐ None of the given options

→

Correct answer : The process of transferring data from the accumulator to memory is called a memory write operation.

Explanation : When data is transferred from the CPU's accumulator/register to memory, the operation is called a memory write. A memory read operation is the opposite process, where data is transferred from memory to the CPU.

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Finish Test

4. Computer Science

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PreviousNext

Question 2

Attempted: 0/14

Revisit Later

What will be the output of the given code?

```
1 var my_arr = [3, 6, 2, 18];
2 function myFunction (num) {
3     return (num + 5);
4 }
5 document.write (my_arr.map(myFunction));
6
```

Select an option

Clear Response

☐ 3, 6, 2, 18

☐ 8, 11, 7, 23

☐ false

☐ true

Correct answer : 8, 11, 7, 23**Explanation :** The array is [3, 6, 2, 18]. The map() function applies myFunction() to every element, and the function adds 5 to each element.

3 + 5 = 8

6 + 5 = 11

2 + 5 = 7

18 + 5 = 23

Therefore, the output is 8, 11, 7, 23.

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⌚ Test Timer: 00:39:47   Finish Test

4. Computer Science   < 7 8 9 10 11 12 13 14 15 16 > 

Question 12

Assume a retail company notices that queries to search for customers by Email are running very slowly.

Increase the Customers table has over 5 million records. The Email column has a UNIQUE index.

```
SELECT * FROM Customers WHERE Email = 'abc@example.com';
```

Which optimization technique will improve the query performance the most?

Select the correct option from the given choices.



Attempted: 0/14
 Revisit Later

Select an option

 Clear Response

☐ Adding more rows to the table

☐ Creating an index on the Email column

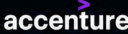
☐ Using a Foreign Key on the Email column

☐ Denormalizing the table

Correct answer : Creating an index on the Email column

Explanation : The query searches the Customers table using the Email column. Since the table contains more than 5 million rows, an index on Email allows the database to locate matching records much faster instead of scanning the entire table.

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Finish Test

4. Computer Science

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Question 7

Attempted: 0/14

Revisit Later

What will be the output of the depicted code?

```
1 class Test {
2     public int x = 45;
3     public int y;
4     public double z;
5     public void f1 (int a) {
6         Y = a;
7     }
8     public double F2() {
9         return 5;
10    }
11 }
12 class T2 {
13     public static void main(String args[]) {
14         Test t = new Test();
15         t.X = 9;
16         t.Y = 18;
17         t.Z = 3;
18         t.f1(21);
19         System.out.println(t.F2());
20     }
21 }
22 }
```

Select an option

Clear Response

☐ 3

☐ 5

☐ 6

☐ 7

Correct answer : 5

Explanation : Initially, $x = 45$, y is declared as an integer, and $z = 3$. The statement $t.f1(21)$ changes y to 21. However, $F2()$ simply returns the constant value 5. Therefore, $\text{System.out.println}(t.F2())$ prints 5.

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Question 11 Attempted: 0/14
Revisit Later

As e-commerce company updates its website weekly by adding new features and fixing bugs using Agile development approach. After each update, the new version is automatically deployed to the live environment. Which key DevOps practice is being implemented to ensure rapid and frequent releases? Analyze the given choices and select the correct option.

←

Select an option Clear Response

☐ The practice involves automating the process of releasing software to production frequently and reliably.

☐ Continuous integration focuses on merging code changes regularly to prevent integration issues.

☐ Continuous testing ensures that automated tests are run continuously to validate code quality.

☐ Monitoring involves tracking the application's performance and user activity after deployment.

Correct answer : The practice involves automating the process of releasing software to production frequently and reliably.

Explanation : The scenario states that after every update, the new version is automatically deployed to the live environment. This describes Continuous Deployment, a DevOps practice in which tested changes are automatically released to production.

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Finish Test

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Question 1

Attempted: 0/14

Revisit Later

You are configuring Amazon S3 bucket-level cross-origin resource sharing and received below bucket policy. What key do you use to generate a CORS preflight request to HTML?

```
1 {
2   "CORSRules": [
3     {
4       "AllowedMethods": ["PUT", "GET"],
5       "AllowedOrigins": true,
6       "AllowedHeaders": ["x-amz", "mal"]
7     }
8   ]
9 }
```

Select an option

Clear Response

☐ corsconfiguration=enabled

☐ collectCoverage=true - corsconfiguration=enabled

☐ coverage - corsconfiguration=enabled

☐ collectCoverage=true - coverageplayertes=enabled

Correct answer : corsconfiguration=enabled

Explanation : Enabling and passing the CORS configuration flag allows Amazon S3 to evaluate cross-origin permissions and respond to browser preflight (OPTIONS) requests.

Q10.

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Question 10

Attempted: 0/14

Revisit Later

Assume a development team is designing a new RESTful API to manage a product catalog for their e-commerce platform. After reviewing industry best practices and the need to specifically support listing the details of a single product identified by its SKU, the engineer for this operation? Select the correct option from the given choices.

←

Select an option

Clear Response

☐ POST /products/{sku}

☐ GET /products/details/{sku}

☐ PUT /products/{sku}

☐ GET /products/{sku}

Correct answer : GET /products/{sku}

Explanation : In RESTful API design standards, retrieving the details of a single resource is performed using the GET method on the resource path identifying that specific item by its unique identifier (e.g., GET /products/{sku}). Verbs or extra action segments like /details are discouraged in REST endpoint naming.

Q11.



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Finish Test

4. Computer Science

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Question 14

Assume a DevOps engineer notices that a containerized application consumes excessive memory, leading to performance issues. What is the recommended approach to address this anomaly? Select the correct option from the given choices.



Attempted: 0/14

 Revisit Later

Select an option

 Clear Response

☐ Increase the memory limit for the container

☐ Use a larger base image for the container

☐ Reduce the number of replicas of the container

☐ Analyze the application code for memory leaks

Correct answer : Analyze the application code for memory leaks

Explanation : If a containerized application is consuming excessive memory, simply increasing the memory limit does not solve the underlying problem. The appropriate approach is to investigate the application for memory leaks and fix the source of excessive memory consumption.

Q12.

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Question 3Attempted: 3/14
🔖 Revisit Later

What does the following line in the given canvas tag indicate?

```
1 <canvas id="myCanvas">  
2   HTML5 canvas not supported.  
3 </canvas>
```

⬅️

Select an option🗑️ Clear Response

☐ A comment for the user

☐ A comment for incompatible browsers

☐ A comment for Internet Explorer and Opera

☐ All of the given options

Correct answer : A comment for incompatible browsers

Explanation : The text inside a <canvas> element, such as HTML5 Canvas not supported, is fallback content. It is displayed when the browser does not support the <canvas> element. Therefore, it serves as a message/comment for incompatible or unsupported browsers.

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Question 7Attempted: 3/14Revisit Later

Consider the given code. What will be the output of the code? Select the correct answer from the given choices.

```
1 class Children
2 {
3     private final void flipper()
4     {
5         System. Out. println("Children");
6     }
7 }
8
9 public class Parent extends Children
10 {
11     public final void flipper()
12     {
13         System. Out. println("Parent");
14     }
15 }
16 public static void main(String[] args)
17 {
18     New Parent(). flipper ();
19 }
```

Select an optionClear Response

☐ Children

☐ Parent

☐ Children
Parent

☐ Parent
Children

Correct answer : Parent

Explanation : Parent extends Children. The Children class contains a private flipper() method. A private method is not inherited by a subclass, so the flipper() method declared in Parent is a separate method. The call new Parent().flipper() therefore invokes the Parent implementation and prints:

Parent

Q14.

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Question 7 | Attempted: 6/14 | Revisit Later | Select an option | Clear Response

What will be the output of the depicted code?

```
1 class Test {
2     public int X = 45;
3     public int Y;
4     public double Z;
5     public void F1(int a) {
6         Y = a;
7     }
8     public double F2() {
9         return 5;
10    }
11 }
12
13 class T2 {
14     public static void main(String args[]) {
15         Test t = new Test();
16         t.X = 9;
17         t.Y = 18;
18         t.Z = 3;
19         t.F1(21);
20         System.out.println(t.F2());
21     }
22 }
```

☐ 3

☐ 5

☐ 6

☐ 7

Correct answer : 5

Explanation : The code creates an object of class Test, sets:

x = 9

y = 18

z = 3

Then F1(21) changes y to 21. However, F2() returns the constant value 5, independent of the values of x, y, or z. Hence the output is 5.

Q15.

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Question 4 Attempted: 0/14 Revisit Later

You want to create a block that changes its shape to a circular block when you move your mouse over it, and reverts back to a rectangular block when you move your mouse away from it, with a slow-fast-slow animation speed. How will you achieve this goal?

←

Select an option Clear Response

☐

```
1 .rect:hover{
2   border-radius:50px;
3   transition: 0.5s;
4 }
```

☐

```
1 .rect:hover{
2   border-radius:40px;
3   transition: 0.5s;
4 }
```

☐

```
1 .rect:hover{
2   border-radius:30px;
3   transition: 0.5s;
4 }
```

☐

```
1 .rect:hover{
2   border-radius:20px;
3   transition: 0.5s;
4 }
```

Correct answer : (A) using border-radius: 50px with transition: 0.5s

Explanation : border-radius is used to round the corners of an element. A sufficiently large border radius, such as 50px for the dimensions implied in the question, makes the rectangular block circular/rounded. The transition property provides the smooth animation when the mouse enters and leaves the element.

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Question 8

Assume that a java application having a java class name 'TeaMac' that can handle different types of tea. The class have a method to boil tea based on the tea selected. A code is written as below:

```
1 public class TeaMac {
2     public coffee boilTea(ChooseTea choosed) {
3         switch (choosed) {
4             case GREEN_TEA:
5                 return boillgreentea();
6             default:
7                 // default code
8         }
9     }
10
11     public list boilTea(ChooseTea choosed) {
12         list teas = new ArrayList(number);
13         for (int i=0; i<number; i++)
14             teas.add(boilTea(choosed));
15         return teas;
16     }
17 }
```

Which type of polymorphism is followed in the given code snippet? Analyze the given choices and select the correct answer.

Select an option

☐ Static polymorphism

☐ Dynamic polymorphism

☐ Late binding

☐ Method overriding

Correct answer : Static polymorphism

Explanation : Overloading the boilTea method with different return types or parameter signatures within the same class is resolved at compile time, known as static (compile-time) polymorphism.

Q17.

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 Finish Test

4. Computer Science



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Question 9

Attempted: 0/14

 Revisit Later

A mobile application needs to send data to a server using an API. The developer decides to use the POST method for this operation. What is the primary characteristic of the POST method in this context?



Select an option



☐ It retrieves data from the server

☐ It updates existing data on the server

☐ It submits data to be processed by the server

☐ It deletes data from the server

Correct answer : It submits data to be processed by the server

Explanation : The HTTP POST method is primarily used to send data to a server, commonly when creating a new resource or submitting information for processing. It is therefore the appropriate choice when a mobile application needs to send data to the server.

Q18.

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4. Computer Science

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Question 2

Attempted: 0/14
Revisit Later

In the given code, when is myFunction() called?

```
1 <html>
2 <head>
3   <script language="JavaScript">
4     function myFunction() {alert("My Function is called");}
5   </script>
6 </head>
7 <body>
8   
9 </body>
10 </html>
```

Select an option

Clear Response

☐ When the page is closed

☐ When the page is loaded

☐ When the user clicks on esc before the page loads

☐ When the page is refreshed

Correct answer : When the user clicks on Esc before the page loads

Explanation : The onabort event is triggered when loading of an object, such as an image, is aborted before it completes. Pressing Esc while the page/resource is loading can abort the loading operation, triggering the onabort event.

Q19.

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Test Timer: 00:18:45

Finish Test

4. Computer Science

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Question 10

Attempted: 0/14

Revisit Later

Consider that a public-facing API has undergone significant changes, including modifications to existing endpoints and data structures, making it incompatible with older client applications. The development team needs to deploy these changes while ensuring existing integrations work. Which API versioning strategy is generally recommended for handling such breaking changes? Select the correct option from the given choices.

←

Select an option

Clear Response

☐ Make the changes directly to the existing endpoints and update the documentation

☐ Use query parameters to specify the desired version (e.g., /api/resource?version=2)

☐ Include the version number in the URL path (e.g., /api/v2/resource)

☐ Use custom HTTP headers to specify the API version

Correct answer : Include the version number in the URL path (e.g., /api/v2/resource)

Explanation : A common and clear API-versioning strategy is to include the version in the URL path. For example:

/api/v1/products

/api/v2/products

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Question 13

Attempted: 0/14
Revisit Later

Consider that a company wants an automatic log whenever an employee record is deleted.

The schema is:

```
1 CREATE TABLE Employees (  
2   EmpID INT PRIMARY KEY,  
3   Name VARCHAR(50)  
4 );  
  
1 CREATE TABLE AuditLog (  
2   LogID INT PRIMARY KEY AUTO_INCREMENT,  
3   EmpID INT,  
4   DeletedOn DATETIME  
5 );
```

Which trigger definition correctly implements this requirement in MySQL? Analyze the given choices and select the correct option.

Select an option

Clear Response

☐

```
1 CREATE TRIGGER trg_DeleteEmployee  
2 ON Employees  
3 AFTER DELETE  
4 AS  
5   INSERT INTO AuditLog (EmpID, DeletedOn)  
6   SELECT EmpID, GETDATE()  
7   FROM deleted;
```

☐

```
1 CREATE TRIGGER trg_DeleteEmployee  
2 AFTER DELETE  
3 ON Employees  
4 FOR EACH ROW  
5   INSERT INTO AuditLog (EmpID, DeletedOn)  
6   VALUES (OLD.EmpID, NOW());
```

☐

```
1 CREATE TRIGGER trg_DeleteEmployee  
2 ON Employees  
3 FOR DELETE  
4 AS  
5   INSERT INTO AuditLog (EmpID, DeletedOn)  
6   SELECT EmpID, NOW()  
7   FROM inserted;
```

☐

```
1 CREATE TRIGGER trg_DeleteEmployee  
2 AFTER UPDATE  
3 ON Employees  
4 FOR EACH ROW  
5   INSERT INTO AuditLog (EmpID, DeletedOn)  
6   VALUES (NEW.EmpID, NOW());
```

Correct answer : Option 2 — AFTER DELETE ON Employees with OLD.EmpID

Explanation : A trigger that records information whenever an employee is deleted must execute after a DELETE operation. For each deleted row, MySQL provides access to the old values through the OLD keyword.

The correct structure is:

```
AFTER DELETE ON Employees  
FOR EACH ROW  
INSERT INTO AuditLog (EmpID, DeletedOn) VALUES (OLD.EmpID, NOW());
```

Therefore, the second option is correct.

Q21.

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4. Computer Science

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Question 5

Attempted: 0/14

Revisit Later

Which of the following statements is **true** considering the code below?

```
1 class Employee extends Person {  
2     private address;  
3 }
```

←

Select an option

Clear Response

☐ Employee has "is-A" relationship with person and "Has-a" relationship with address.

☐ Employee has "is-A" relationship with address and "Has-a" relationship with person.

☐ Employee has "is-A" relationship with person and address.

☐ Employee has "Has-a" relationship with address and person.

Correct answer : Employee has an "is-a" relationship with Person and a "has-a" relationship with Address.

Explanation : Employee extends Person means that Employee is a Person, which represents an is-a relationship/inheritance.

Address address; means that Employee contains an Address object, which represents a has-a relationship/association.

Therefore:

Employee is-a Person
Employee has-a Address

Q22.

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4. Computer Science1234567810

Question 1
What will be the output of the given code when the user clicks the button?

```
1 <body>
2 <button onclick="myFunction(this);">Test attribute!</button>
3 <script type="text/JavaScript">
4   function myFunction (button) {
5     var newAttr = document.createAttribute ("myAttribute");
6     newAttr.value = "My Value";
7     button.setAttributeNode (newAttr);
8     document.write (button.getAttribute ("myAttribute"));
9   }
10 </script>
11 </body>
```

Select an option
☐ False
☐ Test attribute!
☐ My Value
☐ Nothing will happen

Correct answer : My Value

Explanation : The code creates a new attribute called myAttribute and assigns it the value "My Value":

newAttr.value = "My Value";

The attribute is then added to the button using setAttributeNode(). Finally, getAttribute("myAttribute") retrieves the value of that attribute. Therefore, the output is:

My Value

Q23.

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🔍 ⚙️

Finish Test

4. Computer Science

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Question 14

Attempted: 0/14

🔖 Revisit Later

Consider that a team wants to improve deployment speed and reduce downtime when updating their containerized application. Which practice should the team adopt in this scenario? Select the correct option from the given choices.

⬅️

Select an option

🗑️ Clear Response

☐ Use rolling updates with health checks

☐ Increase the number of replicas

☐ Disable the health checks

☐ Use a larger base image

Correct answer : Use rolling updates with health checks

Explanation : Rolling updates allow an application to be updated gradually instead of stopping all existing instances at once. Health checks ensure that the newly updated instances are healthy before traffic is directed to them. This helps reduce downtime and makes deployments safer.

Q24.

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Campus Hiring - Technical 2025 - New Construct

Test Timer: 00:24:10

Finish Test

4. Computer Science

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Question 3

Which of the following codes is appropriate for autoplay?

←

Attempted: 0/14

Revisit Later

Select an option

Clear Response

☐ 1 <audio autoplay>
2 <source src="test.mp3">
3 <source src="test.ogg">
4 </audio>

☐ 1 <audio autoplay>
2 <source src="test.mp3">
3 <source src="test.ogg">
4 </audio autoplay>

☐ 1 <autoplay>
2 <source src="test.mp3">
3 <source src="test.ogg">
4 </autoplay>

☐ 1 <audio>
2 <controls>
3 <source src="test.mp3">
4 <source src="test.ogg">
5 </controls>
6 </audio>

Correct answer : <audio autoplay> with <source> elements

Explanation : The HTML5 <audio> element supports the autoplay attribute. The <source> elements specify the audio files that the browser can choose from.

The correct structure is:

```
<audio autoplay>  
<source src="test.mp3">  
<source src="test.ogg">  
</audio>
```

Therefore, the first option is correct.

Q25.

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Campus Hiring - Technical 2025 - New Construct

Test Timer: 00:15:00

Finish Test

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Attempted: 0/14

Revisit Later

Question 6

Your organization has recently decided to outsource your antivirus management on all systems and servers to a third party for cost saving and workforce contract reasons. Which of the following tasks **cannot** be assigned to the third party?

←

Select an option

Clear Response

☐ Monitoring of servers on your organization's premises

☐ Preparing the internal antivirus policy for the systems

☐ Rectification of virus infection on systems

☐ Updating the definition files as and when they are published on user systems

☐ None of the given options

Correct answer : Preparing the internal antivirus policy for the systems

Explanation : Technical operational activities such as monitoring servers, removing infections, and updating antivirus definition files can generally be outsourced to a third-party security provider.

However, developing the organization's internal security/antivirus policy is a governance responsibility that should remain with the organization because it defines its internal security requirements and policies.

Q26. During deployment, assume a DevOps engineer observes that a containerized application experiences high latency. What approach should the engineer take first to rectify the issue? Select the correct option from the given choices.

- A. Increase the CPU and memory limits
- B. Analyze the network traffic and logs
- C. Restart the container
- D. Reduce the number of containers

Correct answer : B. Analyze the network traffic and logs

Explanation :

The first step in troubleshooting latency is diagnosis—analyzing logs and network metrics pinpoints the root cause (e.g., bottlenecks, network delays, or errors) before making arbitrary configuration or resource changes.

Q27. Consider that two transactions run concurrently. Transaction A: UPDATE Products SET Stock = Stock - 1 WHERE ProductID = 1001; Transaction B: UPDATE Products SET Stock = Stock - 2 WHERE ProductID = 1001; Both succeed, but the final stock shows only the result of one update instead of both. Which concurrency issue is this?

- A. Phantom Read
- B. Lost Update
- C. Non-repeatable Read
- D. Deadlock

Correct answer : B. Lost Update

Explanation :

A Lost Update occurs when two concurrent transactions read and modify the same data item simultaneously, causing the second transaction's write to overwrite and erase the changes made by the first without taking them into account.

Q28. Assume that a company maintains an employee database. Each employee is identified by a unique EmployeeID, but sometimes, employees may change their PhoneNumber. While designing the Employee table, the developer is confused about which column should be chosen as the Primary Key. Which column is the correct choice for the Primary Key? Select the correct option from the given choices.

- A. Name
- B. PhoneNumber
- C. EmployeeID
- D. Department

Correct answer : C. EmployeeID

Explanation :

A Primary Key requires unique, non-null, and immutable values. EmployeeID serves as a permanent unique identifier, whereas PhoneNumber is subject to changes and Name/Department are non-unique.

Q29. During the development of an application, a developer encounters an error message indicating that the API endpoint they are trying to access is not found. Which HTTP status code is most likely associated with this error?

- A. 200
- B. 404
- C. 500
- D. 403

Correct answer : B. 404

Explanation :

HTTP 404 Not Found indicates that the server cannot locate the requested resource or API endpoint. 200 means Success, 500 indicates an Internal Server Error, and 403 indicates Forbidden access.

Q30. In a Web filtering service, the firewall stops the user's request from leaving the network. It then queries the Web filter and, based on the reply from the web filter, it allows or disallows the request. Which of the following is not a disadvantage of using this service?

- A. The Firewall System has to ensure that all requests made by users are copied to the web filter system and hence causing excess use of resources
- B. There is a trust relationship which exists between the firewall system and the web filter and any malicious activity on one system is likely to affect the other system also
- C. In case the web filter starts failing, the network connectivity of the system is tampered with because of the trust relationship that exists between web filters and firewall
- D. A and C

Correct answer : B. There is a trust relationship which exists between the firewall system and the web filter and any malicious activity on one system is likely to affect the other system also

Explanation :

- A and C describe direct operational disadvantages: processing/resource overhead from duplicating requests to the filter and network disruption/latency if the filter fails or times out.
- B represents the standard architectural integration and operational security posture rather than an inherent functional disadvantage of this filtering mechanism.